

MSc INDIVIDUAL PROJECT

MSc HUMAN AND BIOLOGICAL ROBOTICS

IMPERIAL COLLEGE LONDON

DEPARTMENT OF BIOENGINEERING

**Multimodal Teleoperation of a Wearable Dual-Arm
Supernumerary Robotic System: Master Interface
Design, Safety Architecture, and a Workspace
Characterisation**

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CONFIRM: title with Dr. Zhang. Option 2 of three was accepted pending supervisor confirmation; the alternatives and their trade-offs are in `thesis_report/DIVERGENCE.md` Part D.

CONFIRM: word limit and submission format with Dr. Zhang / Prof. Burdet. The 12,000–15,000 word target used here is not authoritative.

FIGURE: Imperial College logo — run `epstopdf title/logo.eps` once ghostscript is available, then restore the `includegraphics` in `title/title.tex`.

Abstract

WRITE: abstract, last, once the Results chapter is final.

Acknowledgements

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Chapter 1

Introduction

Supernumerary robotic limbs (SRLs) are additional arms worn on the body which act alongside the wearer's own limbs rather than replacing or rehabilitating them. They are distinguished from prostheses in that the wearer retains full natural function, and from fixed industrial manipulators in the single respect that matters most for their control: the base of the robot is a person. It accelerates, sways and steps; it is soft, valuable and irreplaceable; and it cannot be commanded to hold still.

The work reported in this thesis concerned the MUlti-limb Virtual Environment (MUVE), known within Imperial College London as the Dr. Octopus platform, which mounts two Kinova Gen3 seven-degree-of-freedom ultralight arms on a backpack frame worn by an operator. The specific problem addressed was the command interface: how a person may direct two redundant seven-degree-of-freedom arms in a manner that is intuitive enough to be usable without training, and safe enough to be acceptable on a robot mounted beside a human head.

The approach taken, following the project's planning report, was a *miniature mannequin master*: a scaled physical replica of the Gen3 kinematic chain, instrumented with potentiometers and an inertial measurement unit, which the operator manipulates with the fingers of one or both hands while the full-scale arms follow. The appeal of this arrangement is that it requires no abstraction on the operator's part. The master has the morphology of the thing it commands, so the mapping from intention to motion is direct rather than learned.

A second architectural property emerged during the work and came to shape the thesis. Because the arms are worn on a backpack, the natural deployment is one in which the arms are borne by one person and commanded by another. That configuration has a direct predecessor in the Fusion system, which mounted a robotic head and two arms on a worn frame and drove them from a remote operator. It is also the configuration that makes the control problem genuinely difficult, because the person issuing commands has neither proprioception of the wearer nor any vestibular sense of the base motion against which the arms must work.

1.1 Objectives

The objectives stated in the project's planning report were framed around a haptic master–slave system with physical springs at the master and virtual spring impedance at the slave. The delivered system diverged from that plan in several respects, for reasons set out in [Chapter 6](#) and analysed in full in the accompanying divergence record. The objectives below are the revised set, aligned to what was built and measured.

1. To design and fabricate a miniature instrumented mannequin master replicating the Gen3 morphology at reduced scale, and to characterise its sensing.
2. To derive and implement the mapping from master pose to slave end-effector command, including the

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coordinate-convention conversion required between the manufacturer’s controller and the robot middleware.

3. To implement a safety architecture appropriate to a manipulator mounted beside a person, and to validate it by systematic fault injection rather than by inspection.
4. To characterise the reachable workspace of the mounted configuration and to determine which manipulation tasks it can support.
5. To establish a verification methodology yielding traceable evidence for every claim, and to apply it across the task set.
6. To identify, by measurement, the constraints that must be resolved before hardware validation and a user study become possible.

1.2 Challenges

Four difficulties consumed the majority of the effort, and each is treated at length in the chapters that follow.

The first was *kinematic dissimilarity between master and slave*. The master replicates the Gen3’s joint sequence but not its proportions: its links sum to 0.272 m against the full-scale arm’s 0.902 m reach. A joint-to-joint mapping is therefore unavailable, and a mapping in task space must contend with the fact that the two chains reach different sets of poses. The resolution adopted was to abandon joint correspondence entirely and map displacement of the master’s tip onto displacement of the slave’s end effector about a re-latched anchor, which is described in [Chapter 3](#).

The second was *a coordinate-convention mismatch*. The manufacturer’s controller reports joint positions in degrees over $[0, 360)$ while the middleware uses radians over $(-\pi, \pi]$, and the two conventions must be reconciled on every command. The failure mode is not a small error: an angle of 350° interpreted as $+350$ rather than -10 commands very nearly a full revolution of a joint standing next to a person. The conversion was therefore isolated in a single module and tested at the seams where such mistakes occur.

The third was *calibration without power-on discontinuities*. The potentiometers report absolute values whose meaning depends on how the master was assembled, so a zero reference must be captured and stored. If that reference is taken from a frame that has failed validation, or if the commanded pose is referenced to the origin of the master’s own frame rather than to the operator’s actual hand position, the first command issued after start-up is a step of the master’s entire tip vector.

The fourth, and the one that proved most pervasive, was that *almost every failure in this system is silent*. An arm that has stopped moving looks identical whether it is blocked for a good reason, blocked for a bad one, or receiving no data at all. A substantial part of the engineering effort went into making internal state legible from outside, and a recurring finding of the results is that the measuring instruments failed more often, and more quietly, than the system being measured.

1.3 Contributions

The contributions claimed are limited to those the implementation and its recorded artefacts can support. Each is stated with its maturity: whether it was implemented, verified in simulation, or validated on physical hardware. **No contribution in this thesis was validated on physical hardware**, for reasons given in [Chapter 6](#).

1. **A complete, safety-instrumented teleoperation stack** for a backpack-mounted pair of Gen3 arms, comprising master sensing, a spherical pose decomposition, collision-aware inverse kinematics with redundancy resolution, a geometric clearance margin referenced to the wearer’s body, and a multi-path emergency stop whose blocking behaviour is externally observable. *Implemented; verified in simulation.*
2. **A coordinate-convention conversion layer** that isolates the degree-to-radian and wrapping problem in one auditable place, with a shortest-path joint difference that distinguishes continuous joints from limited ones. *Implemented; unit-tested.*
3. **A workspace characterisation of the mounted configuration**, including the finding that the two arms share no reachable workspace under the as-built mount, which blocks inter-arm handover for geometric reasons that no control or planning change can address, together with a quantified and buildable mount modification that would resolve it. *Verified in simulation; Chapter 5.*
4. **A quantified account of degraded-input teleoperation** — which capabilities survive the loss of individual sensing channels, what each loss costs, and evidence of progressive hardware degradation between two dated recordings. *Measured from recorded hardware data.*
5. **A verification methodology** requiring that an instrument recover a known answer before a surprising measurement taken with it is reported, applied across a set of recorded runs covering every task, scenario and autonomy condition with automated checking of object handling, gripper actuation and wearer clearance. *Applied throughout.*

The fifth of these arose from necessity rather than principle. Across the project, unexpected results proved more often to be defects in measurement than in the system, and the discipline of validating an instrument before believing its output caught a substantial number of errors that would otherwise have entered the record. It is, in the author’s assessment, the most transferable outcome of the work.

1.4 Thesis structure

[Chapter 2](#) reviews supernumerary limb design, master–slave teleoperation and the mapping problem between dissimilar morphologies. [Chapter 3](#) describes the system as built, from the master hardware through to the safety architecture. [Chapter 4](#) reports the measured outcomes, organised by validation level and labelled throughout with the strength of evidence behind each. [Chapter 5](#) presents the workspace characterisation separately, because it is a chain of measurements whose consequences determine the shape of the task set. [Chapter 6](#) interprets these results, compares the delivered system against the planning report’s targets, and states the limitations honestly. [Chapter 7](#) restates what was achieved against the objectives and sets out the work that would most improve the system.

Chapter 2

Background

Chapter 3

System Design and Implementation

Chapter 4

Results

Chapter 5

Workspace Characterisation and its Consequences

Chapter 6

Discussion

Chapter 7

Conclusion

Chapter A

Repository Structure

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Chapter B

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